Contributing to a Website with Version Control

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1. History of Version Control

- **1970s–80s, SCCS and RCS:** track single files on one machine; one editor at a time.
- **1990, CVS:** first shared server for whole projects, but commits were not atomic.
- **2000, Subversion:** atomic commits (all changes save together or not at all).
- **2005, Git:** everyone holds the full history, so branching and merging are cheap.
- **2008 onward, GitHub and GitLab:** hosting, SSH access, issues and merge requests.

2. Git Basics

Repository

The project folder plus its full history of changes.

Commit

A saved snapshot of your changes with a message explaining why.

Remote

The shared copy on GitLab that everyone pulls from and pushes to.

3. Getting Started

1. **Email the GitLab owner** to ask for access to the project.
2. **Generate a key:** `ssh-keygen -t ed25519 -C "you@example.com"`
   Press Enter through the prompts to accept the defaults.
3. **Copy the public key:** `cat ~/.ssh/id_ed25519.pub`
4. **Add it to GitLab:** User Settings → **SSH Keys**, paste the key, click **Add key**.
5. **Test the connection:** `ssh -T git@gitlab.com`
6. **Clone the repository:** `git clone git@gitlab.com:team/website.git`, then `cd website`.

Only ever share the *public* key (`.pub`); keep the private key to yourself.

4. The Daily Loop

Pull your branch → Edit → Commit → Merge main → Push

```
git checkout my-branch        # first time: git checkout -b my-branch
git pull origin my-branch     # start from the latest of your branch
# ... edit files, test locally ...
git add .
git commit -m "Describe the change"
git fetch origin
git merge origin/main         # merge main into your branch
git push origin my-branch     # share your branch
```

Pull often, commit small, and never share your private key.

5. Editing and Review Workflow

Edit → Preview → Reviewer → Push

1. **Edit** the files on your branch.
2. **Preview:** the website gives you a link to your particular commit. Copy it and paste it into your browser to see your change.
3. **Select a reviewer:** in GitLab, open your merge request and choose a **Reviewer** from the sidebar.
4. **Push** your branch: `git push origin my-branch`.

6. Running the Site on Localhost

Option A: Terminal

```
# plain HTML site
python3 -m http.server 8000
# http://localhost:8000

# Node-based site
npm install
npm run dev
```

Stop with `Ctrl+C`.

Test on localhost, then commit and push.

Option B: MAMP (PHP/MySQL)

1. Install MAMP.
2. Put the repo in `htdocs`, or set the Document Root in Preferences → Server.
3. Click **Start**.
4. Open `http://localhost:8888`.

If it fails, check Preferences → Ports.

7. Useful Commands

Command	What it does
`git status`	Show changed and staged files
`git diff`	See exactly what you changed
`git log -oneline`	Compact history of commits
`git branch`	List branches (current one marked)
`git checkout -b name`	Create and switch to a new branch
`git fetch origin`	Download updates without merging
`git stash` / `git stash pop`	Shelve unfinished work, then bring it back
`git restore file`	Discard your edits to a file